2014 - 09 iunie

1. Ne interesează o linie (sau un polinom) care să treacă printre punctele date și care să aproximeze funcția cât mai bine.

Ne-am dori $E = E_2(a_0, a_1) = \sum_{i=1}^{n} [y_i - (a_1 x_i + a_0)]^2$ să fie cât mai mică

2. U = diag(x) * ones(n) - ones(n) * diag(x) + eye(n)

V = (a - diag(x)) * eye(n) + eye(n)

L = [l1 l2 ... ln] - prod(V) ./ prod(U);

Voalorea în a =

$$b = \sum_{i=1}^{n} l_i \cdot y_i = prod(V) \cdot / prod(U) * y_i$$

3. FFT calculează DFT a unei secvențe sau IFT.

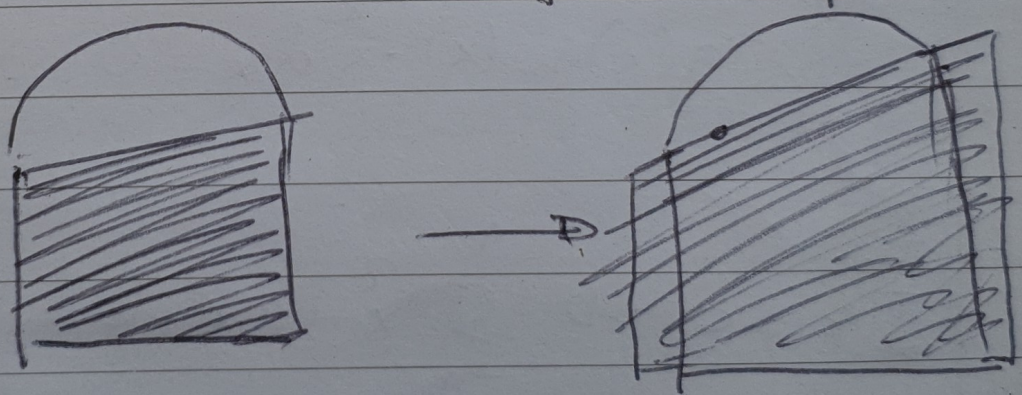
Dom timp/spațiu $\xrightleftharpoons{\mathcal{F}}$ Frecvență

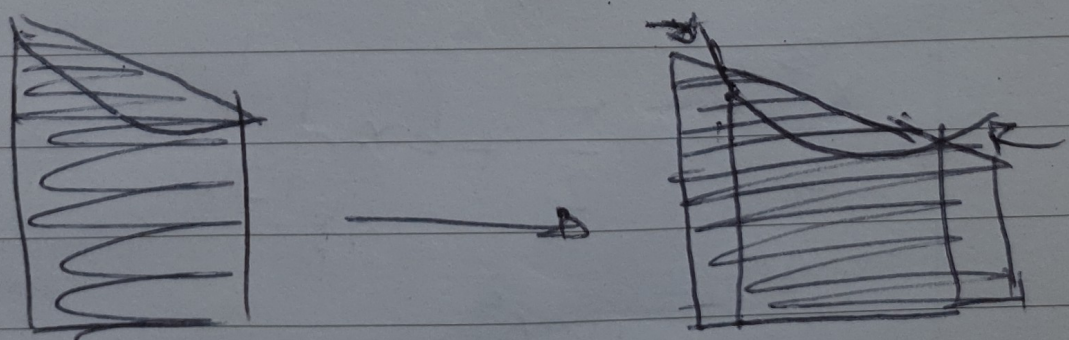
FFT grupează termenii de la

DFT mk-un produs ~~de factori~~ astfel mai reduce complexitatea de la $O(n^2)$ la $O(n \log n)$

Importanta de viteză e atenuată în zilele noastre considerăm FFT cel mai important algoritm din istorie.

4. NC ~~Newton-Cotes~~ - exacte pt. pol. de grad cel mult n
   IG - extindere la $2n-1$ (g.d.u)
   - alege punctele optim





- minimizează eroarea:

$$\int_a^b f(x)dx \approx \sum_{i=1}^{n} c_i f(x_i)$$

5. $\int_{-1}^{1} f(x)dx \approx a_1 f(x_1) + a_2 f(x_2)$

$\int_a^b g(x)w(x)dx \approx \sum_{i=1}^{N} A_i \cdot N \quad f(x_i)$

$g(x) = \prod_{i=1}^{2} (x - x_i)$

$w(x) = 1$

$[a; b] = [-1; 1]$

$g(x) = (x - x_1)(x - x_2) = x^2 + ax + b$

$\int_{-1}^{1} g(x)w(x) \cdot x^k dx = 0 \quad \forall k = \overline{0,1}$

(I) $k = 0 \Rightarrow \int_{-1}^{1} x^2 + ax + b\,dx = 0$

$\frac{x^3}{3} + a\frac{x^2}{2} + bx \Big|_{-1}^{1} = 0$

$\frac{x^3}{3} + a\frac{x^2}{2}$

$\frac{1}{3} + a\frac{1}{2} + b - \frac{-1}{3} - a\frac{x}{2} - (-b) =$

$\frac{2}{3} + 2b = 0 \Rightarrow b = -\frac{2}{3} \cdot \frac{1}{2} = -\frac{1}{3}$

(II) $k = 1 \Rightarrow \int_{-1}^{1} x^3 + ax^2 + bx\,dx = 0$

$\frac{x^4}{4} + a\frac{x^3}{3} + \frac{bx^2}{2}\Big|_{-1}^{1} = 0$

$\frac{1}{4} + a \cdot \frac{1}{3} + \frac{b}{2} - \frac{1}{4} - a\frac{(-1)}{3} - \frac{b}{2} = 0$

$\frac{2}{3}a = 0 \Rightarrow a = 0$

$\Rightarrow g(x) = x^2 - \frac{1}{3} = (x - \sqrt{\frac{1}{3}})(x + \sqrt{\frac{1}{3}})$

Integrala trebuie să fie exactă pt:

(I) $f(x) = 1 \Rightarrow$

$\int_{-1}^{1} 1\,dx = a_1 + a_2 = 2$

(II) $f(x) = x \Rightarrow$

$\int_{-1}^{1} x\,dx = \frac{x^2}{2}\Big|_{-1}^{1} = \frac{1}{2} - \frac{1}{2} = 0 = \sqrt{\frac{1}{3}}a_1 - \sqrt{\frac{1}{3}}a_2$

GV: $2 \cdot 2 - 1 = 3$

$$\left.\begin{array}{l} q_1 + q_2 = 2 \\ \sqrt{\frac{1}{3}} q_1 - \sqrt{\frac{1}{3}} q_2 = 0 \end{array}\right\} \Rightarrow q_1 = q_2 = 1$$

$$\Rightarrow \int_{-1}^{1} f(x)\,dx \cong 1 \cdot f\left(\sqrt{\frac{1}{3}}\right) + 1 \cdot f\left(-\sqrt{\frac{1}{3}}\right)$$

6. $2n+1$ componente
?
0

7. ✓

8. $A = \begin{bmatrix} 3 & 1 & 0 \\ 1 & 3 & 1 \\ 0 & 1 & 3 \end{bmatrix}$

$$E^{(1)} = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$$

$$\det(\lambda I - E^{(1)}) = 0$$

$$\begin{vmatrix} 3-\lambda & 1 \\ 1 & 3-\lambda \end{vmatrix} = 9 - 6\lambda + \lambda^2 - 1$$

$$= \lambda^2 - 6\lambda + 8 = (\lambda - 4)(\lambda - 2)$$

$$\Rightarrow \begin{cases} \lambda_1 = 2 \\ \lambda_2 = 4 \end{cases}$$

$$\Rightarrow \sqrt{1} = 2$$

$$B_1 = A^{(1)} - \sqrt{1}\, I_3 = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

$$G_{1,2} = \begin{bmatrix} c & -s & 0 \\ s & c & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

$$\rho = \sqrt{1 + 1} = \sqrt{2}$$

$$s = -\frac{1}{\sqrt{2}}$$

$$c = \frac{1}{\sqrt{2}}$$

$$G_{12} \cdot B_1 = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ -1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix} = \begin{bmatrix} \sqrt{2} & \sqrt{2} & 1/\sqrt{2} \\ 0 & 0 & 1/\sqrt{2} \\ 0 & (1) & 1 \end{bmatrix}$$

$$G_{23} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & c & -s \\ 0 & s & c \end{bmatrix} \quad \begin{matrix} s = \sqrt{1} \\ s = -1 \\ c = 0 \end{matrix} \longrightarrow$$

$$G_{23} \cdot G_{12} \cdot B_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} \sqrt{2} & \sqrt{2} & 1/\sqrt{2} \\ 0 & 0 & 1/\sqrt{2} \\ 0 & 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} \sqrt{2} & \sqrt{2} & 1/\sqrt{2} \\ 0 & 1 & 1 \\ 0 & 0 & -1/\sqrt{2} \end{bmatrix} = R^{(1)}$$

$$\underbrace{G_{23} \cdot G_{12}}_{Q^{(1)t}} \cdot B_1 = R^{(1)}$$

$$Q^{(1)} = G_{12}^{\,t} \, G_{23}^{\,t} =$$

$$= \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} & 0 \\ 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1/\sqrt{2} & 0 & 1/\sqrt{2} \\ 1/\sqrt{2} & 0 & -1/\sqrt{2} \\ 0 & 1 & 0 \end{bmatrix} \longrightarrow$$

$$A^{(2)} = R^{(1)} Q^{(1)} + \sigma_1 I_3$$

$$A^{(2)} = \begin{bmatrix} \sqrt{2} & \sqrt{2} & 1/\sqrt{2} \\ 0 & 1 & 1 \\ 0 & 0 & -1/\sqrt{2} \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} & 0 & 1/\sqrt{2} \\ 1/\sqrt{2} & 0 & -1/\sqrt{2} \\ 0 & 1 & 0 \end{bmatrix}$$

$$+\sigma_1 I_3 = \begin{bmatrix} 2 & 1/\sqrt{2} & 0 \\ 1/\sqrt{2} & 1 & -1/\sqrt{2} \\ 0 & (-1/\sqrt{2}) & 0 \end{bmatrix} + \sigma_1 I_3$$

Nel $e \neq 0 \Rightarrow$ continua

Aprox. funcții continue în sensul cmmp

$$S_n(x) = \frac{a_0}{2} + \sum_{k=1}^{n} (a_k \cos kx + b_k \cos kx)$$

$$a_k = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos kx \, dx$$

Ex: $f(x) = \begin{cases} 0, & -\pi < x \leq 0 \\ 1, & 0 < x < \pi \end{cases}$

$$\boxed{a_0} = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x)\,dx = \frac{1}{\pi} \int_{-\pi}^{0} 0\,dx + \frac{1}{\pi} \int_{0}^{\pi} 1\,dx = \boxed{1}$$

$$a_k = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos kx \, dx$$

$$= \frac{1}{\pi} \int_{-\pi}^{0} 0 \cdot \cos kx \, dx + \frac{1}{\pi} \int_{0}^{\pi} 1 \cdot \cos kx \, dx$$

$$= \frac{1}{\pi} \left. \frac{\sin kx}{k} \right|_0^{\pi} = 0$$

$$b_k = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin kx \, dx$$

$$= \frac{1}{\pi} \int_{-\pi}^{0} 0 \cdot \sin kx \, dx + \frac{1}{\pi} \int_{0}^{\pi} 1 \cdot \sin kx \, dx$$

$$= -\frac{1}{\pi} \left. \frac{\cos kx}{k} \right|_0^{\pi} = -\frac{1}{\pi} \left(\frac{(-1)^k - 1}{k} \right)$$

$$= \frac{1}{\pi} \left(\frac{1 - (-1)^k}{k} \right)$$

$$S_n(x) = \frac{1}{2} + \frac{1}{\pi} \sum_{k=1}^{n} \left(\frac{1 - (-1)^k}{k} \right) \sin kx$$

2013 - 14 iunie - A

1. Clasa + gradul de derivab. / continuitate

C_1, C_2 + poveste?